Conjugate fibres and irrationality exponents

Christopher D. Long
galizur@gmail.com

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Abstract

Let $\xi \in \mathbb{R}^\times$ and let $a, b \in \overline{\mathbb{Q}}^\times$ satisfy $e^{a\xi} = b$. Put $F = \mathbb{Q}(a, b)$, let $d = [F : \mathbb{Q}]$, and let r be the number of embeddings $\sigma : F \hookrightarrow \mathbb{C}$ for which the conjugate relation

$$e^{\sigma(a)\xi} = \sigma(b)$$

still holds. We prove

$$\mu(\xi) \leq \frac{2d}{r}.$$

The proof combines the separated-weight interpolation determinant of Liu–Jiang–Zhang with the product formula. At each compatible embedding the rational approximation to ξ produces the full analytic determinant saving; at the remaining embeddings a crude holomorphic estimate suffices. We then optimize over all algebraic exponential certificates, over powers of a fixed certificate, and over the lattice generated by several certificates. In particular, full compatibility after taking a power implies $\mu(\xi) = 2$, and for every positive algebraic $\alpha \neq 1$ we obtain

$$\mu(\log \alpha) \leq 2 \min_{N \geq 1} [\mathbb{Q}(\alpha^N) : \mathbb{Q}].$$

1 Introduction

For an irrational real number x, its *irrationality exponent* is

$$\mu(x) = \sup \left\{ \nu > 0 : 0 < \left| x - \frac{p}{q} \right| < q^{-\nu} \text{ for infinitely many } p \in \mathbb{Z},\ q \in \mathbb{Z}_{>0} \right\}.$$

Dirichlet's theorem gives $\mu(x) \geq 2$ for every irrational x; see, for example, [2, Chapter I]. Roth's theorem gives equality for every irrational algebraic number [8]. For explicit transcendental constants, proving the same exponent is generally much harder.

A separated-weight interpolation determinant was recently used to prove $\mu(\pi) = 2$ [7]. Liu, Jiang, and Zhang then developed a moving-centre form of the interpolation theorem and proved

$$\mu(\log q) = 2 \qquad (q \in \mathbb{Q}_{>0} \setminus \{1\})$$

[5]. Their paper explicitly points out the next arithmetic obstruction: for an algebraic argument the cleared determinant is an algebraic integer, so a lower bound at one embedding must be replaced by estimates for its conjugates. The purpose of this note is to carry out that norm argument when a specified collection of conjugate exponential fibres remains compatible.

Fix $\xi \in \mathbb{R}^\times$ and $a, b \in \overline{\mathbb{Q}}^\times$ with $e^{a\xi} = b$. Let

$$F = \mathbb{Q}(a, b), \qquad d = [F : \mathbb{Q}], \qquad \Sigma_F = \mathrm{Hom}_{\mathbb{Q}}(F, \mathbb{C}),$$

and define the set of compatible conjugate fibres by

$$\Sigma_\xi(a, b) = \left\{ \sigma \in \Sigma_F : e^{\sigma(a)\xi} = \sigma(b) \right\}. \tag{1.1}$$

The distinguished inclusion $F \subset \mathbb{C}$ belongs to $\Sigma_\xi(a, b)$, so $1 \le r := |\Sigma_\xi(a, b)| \le d$.

Theorem 1.1 (Conjugate-fibre theorem)**.** *Let $\xi \in \mathbb{R}^\times$ and $a, b \in \overline{\mathbb{Q}}^\times$ satisfy $e^{a\xi} = b$. With d and r as above, for every real number*

$$\nu > \frac{2d}{r}$$

there is $Q = Q(a, b, \xi, \nu)$ such that

$$\left| \xi - \frac{p}{q} \right| \ge q^{-\nu} \qquad (p \in \mathbb{Z},\ q \in \mathbb{Z}_{>0},\ q \ge Q).$$

Consequently

$$\boxed{\mu(\xi) \le \frac{2d}{r}}.$$

If every embedding is compatible, so that $r = d$, then

$$\boxed{\mu(\xi) = 2}.$$

The last assertion uses only Dirichlet's theorem and the fact that ξ is transcendental. Indeed, if ξ were a nonzero algebraic number, then $a\xi$ would be nonzero algebraic, whereas $e^{a\xi} = b$ is algebraic, contradicting the Hermite–Lindemann theorem [1, Chapter 1].

The proof has three inputs. First, the bounded-fibre interpolation theorem of Liu–Jiang–Zhang produces a nonzero full-row minor for the centres

$$(b^j, jap_1/q_1, \ldots, jap_m/q_m), \qquad 0 \le j < K.$$

This covers both distinct multiplicative coordinates and roots of unity. Second, at a compatible embedding σ, the identity

$$e^{j\sigma(a)\xi} = \sigma(b)^j$$

permits the same exact row translation and Taylor-collision estimate as in [5]. Third, after a uniform denominator clearing, the determinant is a nonzero algebraic integer. The norm lower bound therefore compares the strong estimates at the r compatible embeddings with crude growth estimates at the other $d - r$ embeddings. The two leading terms are respectively $r\nu$ and d; the sharp parameter threshold for this scalar norm construction is $2d/r$.

Sections 2–5 prove Theorem 1.1. Section 6 gives the optimization, power-saturation, torsion-defect, and relation-lattice consequences.

2 The interpolation input

We recall the precise form of the geometric input that is needed. The following is [5, Theorem 9]. Fix positive rational weights

$$W = (w_0, w_1, \dots, w_m), \qquad V = (v_0, w_1/\theta, \dots, w_m/\theta), \qquad 0 < \theta < 1.$$

Let $\mathcal{P}_W(H)$ be the span of the monomials $Y^h X^\alpha$ with

$$w_0 h + \sum_{i=1}^{m} w_i \alpha_i \le H,$$

and let $\mathcal{J}_V(H)$ be the quotient of $\mathbb{C}[[t, u_1, \dots, u_m]]$ retaining precisely the monomials $t^s u^\beta$ with

$$v_0 s + \frac{1}{\theta} \sum_{i=1}^{m} w_i \beta_i < H.$$

Theorem 2.1 (Bounded-fibre moving-centre interpolation)**.** *Fix $m, K \ge 1$, $1 \le \kappa \le K$, positive rational w_0, v_0, and $0 < \theta < 1$ such that*

$$K \left(\frac{w_0}{v_0}\right) \theta^m < 1, \qquad \kappa \theta^m < 1. \tag{2.1}$$

There are successive lower thresholds for the positive rational weights $w_1, \dots, w_m$, independent of the interpolation centres, with the following property. Let

$$(y_j, c_{j1}, \dots, c_{jm}) \in \mathbb{C}^\times \times \mathbb{C}^m, \qquad 0 \le j < K,$$

assume that at most κ indices have a given value of y_j, and assume that

$$j \ne \ell,\ y_j = y_\ell \quad \Longrightarrow \quad c_{ji} \ne c_{\ell i} \quad (1 \le i \le m). \tag{2.2}$$

Then, for every sufficiently large admissible degree H, the map

$$\mathcal{P}_W(H) \longrightarrow \bigoplus_{j=0}^{K-1} \mathcal{J}_V(H)$$

obtained by substituting

$$Y = y_j(1 + t), \qquad X_i = c_{ji} + u_i + \log(1 + t)$$

is onto. The divisibility modulus for H depends only on the weights; the eventual threshold may depend on the centres.

The proof in [5] uses a local intersection-length estimate, separated weights, a curve inequality, an ordinary blow-up, and Serre vanishing. We shall also use the determinant translation and collision estimates from Sections 4–6 of that paper. Those estimates are uniform over the selected full-row minor and its columns.

2.1 Parameters

The following elementary lemma isolates the only change from the parameter choice for a single rational logarithm.

Lemma 2.2 (Gap parameters)**.** *Let d, r be positive integers and let $\nu > 2d/r$. There exist rational numbers*

$$0 < \theta < A < B < 1, \qquad 0 < \eta < 1,$$

and rational numbers $C > 1$ such that

$$A^2 < \theta < A, \tag{2.3}$$

$$g := r\nu\big(A(1-\eta) - \theta\big) - d(1-\theta) > 0, \tag{2.4}$$

$$\frac{A}{\theta} < C < \frac{1}{A}, \qquad A < B < \min\{C^{-1}, C\theta\}. \tag{2.5}$$

In particular $C\theta < 1$, $CB < 1$, $C\theta/B > 1$, and $B/A > 1$.

Proof. Choose a rational number

$$\frac{1}{2} < \rho < 1 - \frac{d}{r\nu};$$

this interval is nonempty because $r\nu > 2d$. For a sufficiently small positive rational δ, put

$$\theta = 1 - \delta, \qquad A = 1 - \rho\delta.$$

Then $A > \theta$ and

$$\theta - A^2 = (2\rho - 1)\delta - \rho^2\delta^2 > 0.$$

Moreover

$$r\nu(A - \theta) - d(1 - \theta) = \delta\big(r\nu(1 - \rho) - d\big) > 0.$$

Choose a sufficiently small positive rational η to preserve this strict inequality after replacing A by $A(1 - \eta)$. Since $A^2 < \theta$, the interval $(A/\theta, 1/A)$ is nonempty; choose rational C in it. Both C^{-1} and $C\theta$ exceed A, so a rational B satisfying (2.5) exists. Finally $C\theta < \theta/A < 1$. □

For a positive integer m, set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{2.6}$$

Then

$$K\left(\frac{w_0}{v_0}\right)\theta^m = \frac{1}{2}, \qquad K\theta^m \longrightarrow 0, \qquad \frac{K}{w_0} \longrightarrow 0, \qquad v_0 \longrightarrow \infty. \tag{2.7}$$

For later use put

$$L = \frac{\eta^2 K\theta^m}{(m+1)v_0 A^m} = \frac{\eta^2}{2(m+1)}\left(\frac{B}{A}\right)^m. \tag{2.8}$$

Thus $L \to \infty$ as $m \to \infty$.

3 The number-field determinant

Fix the data of Theorem 1.1, and suppose for contradiction that there is a real $\nu > 2d/r$ for which

$$\left|\xi - \frac{p}{q}\right| \le q^{-\nu} \tag{3.1}$$

holds for arbitrarily large positive integers q. Replacing ν by a slightly smaller rational number still larger than $2d/r$, we may assume that ν is rational.

Choose the parameters of Lemma 2.2. Fix a rational truncation parameter

$$\Phi > \frac{2}{\theta}. \tag{3.2}$$

We shall first take m large, then choose m approximations

$$x_i = \frac{p_i}{q_i}, \qquad |\xi - x_i| \le q_i^{-\nu}, \qquad w_i = \lceil \log q_i \rceil, \qquad 1 \le i \le m,$$

successively. The denominators are taken large enough that $x_i \neq 0$, $|x_i| \le |\xi| + 1$, and the weights meet the successive thresholds in Theorem 2.1. This is possible because those thresholds are independent of the centres.

Use the centres

$$y_j = b^j, \qquad c_{ji} = jax_i, \qquad 0 \le j < K, \quad 1 \le i \le m. \tag{3.3}$$

We apply Theorem 2.1 with $\kappa = K$. The second inequality in (2.1) holds for large m by $K\theta^m \to 0$. If $y_j = y_\ell$ and $j \neq \ell$, then

$$c_{ji} - c_{\ell i} = (j - \ell)ax_i \neq 0$$

for every i, so (2.2) holds. Thus the same argument covers roots of unity and non-torsion values of b.

Set

$$T_i = \left\lceil \frac{\Phi w_i}{v_0} \right\rceil, \qquad G_i(t) = \sum_{1 \le k < T_i} \frac{(-1)^{k+1}}{k} t^k, \qquad L_i = \operatorname{lcm}(1, \dots, T_i - 1), \tag{3.4}$$

where an empty sum and empty least common multiple are interpreted as 0 and 1. Because $v_0 T_i > w_i/\theta$, replacing $\log(1+t)$ by $G_i(t)$ is an invertible filtered change on every retained jet packet [5, Section 4].

For an admissible degree H, use the monomials

$$P = Y^h X^\alpha, \qquad w_0 h + \sum_i w_i \alpha_i \le H,$$

as columns, and use the triples

$$\varrho = (j, s, \beta), \qquad 0 \le j < K, \qquad v_0 s + \frac{1}{\theta} \sum_i w_i \beta_i < H,$$

as rows. The matrix entry is

$$[t^s u^\beta]\, b^{jh} (1+t)^h \prod_{i=1}^{m} (jax_i + G_i(t) + u_i)^{\alpha_i}. \tag{3.5}$$

For every sufficiently large admissible H, interpolation supplies a nonzero square minor $\Delta_H \in F^\times$ containing every row. Let $M = M_H$ be its size, and put

$$\bar{\beta}_H = \frac{1}{MH} \sum_{\varrho=(j,s,\beta)} \sum_i w_i \beta_i. \tag{3.6}$$

Then

$$M \sim \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}, \qquad 0 \le \bar{\beta}_H \le \theta. \tag{3.7}$$

3.1 Algebraic-integral clearing and the norm inequality

Choose an integer $D \ge 1$ such that

$$Da, \ Db \in \mathcal{O}_F.$$

Put

$$\mathfrak{D}_H = \prod_{i=1}^{m} L_i^{\lfloor H/w_i \rfloor}.$$

Scale the column (h, α) and the row (j, s, β) by

$$C_{h,\alpha} = D^{(K-1)h} \prod_i (Dq_i)^{\alpha_i}, \qquad S_{j,s,\beta} = \mathfrak{D}_H \prod_i (Dq_i)^{-\beta_i}. \tag{3.8}$$

Lemma 3.1 (Norm lower bound)**.** *Let $\Lambda = 4 \log 2$ and $w_* = \min_i w_i$. Define*

$$E_{\mathrm{ar}} = \frac{\Lambda \Phi m}{v_0} + \Lambda \sum_{i=1}^{m} \frac{1}{w_i} + \frac{\theta + \log D}{w_*} + \frac{(K-1)\log D}{w_0}. \tag{3.9}$$

Then

$$0 \le \sum_{\sigma \in \Sigma_F} \frac{\log |\sigma(\Delta_H)|}{MH} + d(1 - \bar{\beta}_H + E_{\mathrm{ar}}). \tag{3.10}$$

Proof. The entry obtained from (3.5) after the row and column scalings is zero unless $\beta \le \alpha$. Otherwise it equals

$$\mathfrak{D}_H (Db)^{jh} D^{(K-1-j)h} \binom{\alpha}{\beta} [t^s](1+t)^h \cdot \prod_{i=1}^{m} \big(j(Da)p_i + Dq_i G_i(t)\big)^{\alpha_i - \beta_i}.$$

The denominator contributed by the ith truncated logarithm divides $L_i^{\alpha_i - \beta_i}$, and $\alpha_i - \beta_i \le \lfloor H/w_i \rfloor$. Hence every scaled entry belongs to $\mathcal{O}_F$. The scaled determinant

$$\widetilde{\Delta}_H = \Delta_H \prod_{\text{columns}} C_{h,\alpha} \prod_{\text{rows}} S_{j,s,\beta}$$

is therefore a nonzero algebraic integer. Consequently

$$1 \le |\,\mathrm{N}_{F/\mathbb{Q}}(\widetilde{\Delta}_H)| = \prod_{\sigma \in \Sigma_F} |\sigma(\widetilde{\Delta}_H)|;$$

we count complex embeddings separately, as in the usual product formula [6, Chapter I].

It remains to bound the positive rational scaling factor. As in [5, Section 4],

$$\frac{\log \mathfrak{D}_H}{H} \le \frac{\Lambda \Phi m}{v_0} + \Lambda \sum_i \frac{1}{w_i}.$$

Every selected column satisfies

$$\sum_i \alpha_i \log q_i \le H, \qquad |\alpha| \le \frac{H}{w_*}, \qquad h \le \frac{H}{w_0}.$$

Moreover, since $0 \le w_i - \log q_i < 1$,

$$\sum_{\text{rows}} \sum_i \beta_i \log q_i \ge MH\bar{\beta}_H - \frac{MH\theta}{w_*}.$$

The negative row contribution involving $\log D$ may be discarded. Thus the logarithm of the total scaling factor, divided by MH, is at most $1 - \bar{\beta}_H + E_{\mathrm{ar}}$. Taking logarithms of the norm inequality proves (3.10). □

4 Archimedean determinant estimates

Set

$$C_a = \max_{\sigma \in \Sigma_F} \max\{1, |\sigma(a)|\}, \qquad C_b = \max_{\sigma \in \Sigma_F} \max\{1, |\sigma(b)|\}, \qquad X = |\xi| + 1.$$

All constants in this section are uniform over the finite set Σ_F.

4.1 Compatible embeddings

For $\sigma \in \Sigma_\xi(a, b)$, define

$$z_j(t) = j\sigma(a)\xi + \log(1+t), \qquad \epsilon_{ji} = j\sigma(a)(x_i - \xi), \qquad \tau_i(t) = G_i(t) - \log(1+t).$$

Compatibility gives the exact identities

$$e^{z_j(t)} = \sigma(b)^j (1+t), \qquad j\sigma(a)x_i + G_i(t) + u_i = z_j(t) + u_i + \epsilon_{ji} + \tau_i(t). \tag{4.1}$$

For a selected column $P = Y^h X^\alpha$ and a multi-index λ, let

$$f_{\lambda,P}(z) = [u^\lambda] P(e^z, z + u_1, \dots, z + u_m) = \binom{\alpha}{\lambda} e^{hz} z^{|\alpha| - |\lambda|},$$

with the convention that this is zero unless $\lambda \le \alpha$. Expanding (4.1) gives the same exact row translation as in [5, Section 5.1]:

$$\begin{aligned} \mathrm{row}^\sigma_{j,s,\beta} = \sum_{\substack{\lambda \ge \beta \\ w\lambda \le H}} \binom{\lambda}{\beta} \sum_{0 \le e \le \lambda - \beta} \binom{\lambda - \beta}{e} \epsilon_j^{\lambda - \beta - e} \\ \cdot \sum_{k=0}^{s} [t^k] \prod_i \tau_i(t)^{e_i} \left([t^{s-k}] f_{\lambda,P}(z_j(t)) \right)_P. \end{aligned} \tag{4.2}$$

Here and below $w\lambda = \sum_i w_i\lambda_i$.

Since

$$|\epsilon_{ji}| \le KC_a q_i^{-\nu} \le KC_a e^{\nu} e^{-\nu w_i},$$

and a nonzero tail coefficient in (4.2) uses $we < H/\Phi$, every scalar multiplying a translated test row is bounded by

$$\exp\{-\nu w(\lambda - \beta) + HE_{\mathrm{tr}}\}, \tag{4.3}$$

where one may take

$$E_{\mathrm{tr}} = \frac{\nu}{\Phi} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2KC_a) + \nu}{w_*}. \tag{4.4}$$

Choose

$$R = 100K(1 + C_a|\xi|).$$

On $|t| \le 1/2$, every $z_j(t)$ lies in $|z| < R/2$. The Taylor-collision argument of [5, Section 5.2], originally appearing in this interpolation-determinant form in [4], gives the following uniform estimate. Put

$$E_{\mathrm{hol}} = \frac{R}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R)}{w_*}, \qquad c = \frac{\log 2}{4}. \tag{4.5}$$

Let

$$S = \sum_i w_i, \qquad L_H = L\left(1 + \frac{S}{AH}\right)^{-m}.$$

Thus $L_H \to L$ as $H \to \infty$.

Lemma 4.1 (Compatible-place estimate)**.** *There is a quantity $\delta_H \to 0$, independent of $\sigma \in \Sigma_\xi(a,b)$ and of the selected full-row minor, such that*

$$\frac{\log|\sigma(\Delta_H)|}{MH} \le E_{\mathrm{tr}} + E_{\mathrm{hol}} + \delta_H + \max\left\{-cL_H, -\nu(A(1-\eta) - \bar\beta_H)\right\}. \tag{4.6}$$

Proof. We indicate the complete reduction to the cited collision estimate. The row expansion (4.2) has only polynomially many terms in H for each row. Choose one translated row from each original row and let n_λ be the number chosen with transverse index λ. Expanding the entire functions $f_{\lambda,P}$ in Taylor series in z/R, any nonzero determinant contribution uses distinct Taylor degrees within each fixed λ-group. Hence

$$|\det T| \le \exp\left\{-c\sum_\lambda n_\lambda^2 + MH(E_{\mathrm{hol}} + o(1))\right\}.$$

This is exactly the collision estimate of [5, Section 5.2]; its proof is uniform in the columns and depends on the fibre only through the radius R.

Let

$$D_A(H) = \#\{\lambda \in \mathbb{Z}^m_{\ge 0} : w\lambda \le AH\}.$$

The same unit-cube estimates as in [5, Section 6] give

$$M \ge \frac{K\theta^m H^{m+1}}{(m+1)!v_0\prod_i w_i}, \qquad D_A(H) \le \frac{(AH+S)^m}{m!\prod_i w_i}.$$

If at least ηM selected indices have weight at most AH, then Cauchy–Schwarz yields

$$\frac{1}{MH}\sum_\lambda n_\lambda^2 \ge L_H.$$

Otherwise more than $(1-\eta)M$ indices have weight greater than AH. Combining this with the scalar estimate (4.3) gives the exponent $-\nu(A(1-\eta)-\bar{\beta}_H)$. The number of multilinear expansion terms is polynomial to the power M; after division by MH its contribution is $o(1)$. This proves (4.6). □

4.2 The remaining embeddings

At an incompatible embedding no approximation gain is available, but none is needed. A direct Cauchy estimate gives a normalized cost that can be made arbitrarily small by the separated parameter choices.

Lemma 4.2 (Crude estimate at every embedding)**.** *For every $\sigma \in \Sigma_F$,*

$$\frac{\log|\sigma(\Delta_H)|}{MH} \le E_{\mathrm{bad}} + \frac{\log M}{2H}, \tag{4.7}$$

where

$$E_{\mathrm{bad}} = \frac{K\log C_b + \log(3/2)}{w_0} + \frac{\log\bigl(2(1+KC_aX+\log 2)\bigr)}{w_*} + \frac{\log 2}{v_0}. \tag{4.8}$$

Proof. The conjugate of (3.5) is zero unless $\beta \le \alpha$. Otherwise it is

$$\sigma(b)^{jh}\binom{\alpha}{\beta}[t^s](1+t)^h \prod_i (j\sigma(a)x_i + G_i(t))^{\alpha_i-\beta_i}.$$

On $|t| = 1/2$ we have $|G_i(t)| \le \log 2$, while $|x_i| \le X$, $j < K$, $s \le H/v_0$, $h \le H/w_0$, and $|\alpha| \le H/w_*$. Cauchy's estimate and $\binom{\alpha}{\beta} \le 2^{|\alpha|}$ therefore bound each entry by $\exp(HE_{\mathrm{bad}})$. Hadamard's inequality contributes at most $M^{M/2}$, which gives (4.7). □

4.3 Making the errors small

The choices are made in the following order. First choose Φ so large that the contribution $r\nu/\Phi$ is smaller than any prescribed positive tolerance. Next let m tend to infinity. By (2.7) and (2.8),

$$\frac{m}{v_0} \to 0, \qquad \frac{K}{w_0} \to 0, \qquad \frac{1}{v_0} \to 0, \qquad L \to \infty.$$

The constants D, C_a, C_b, ξ, d, r are fixed. Thus m may be chosen so that all terms in (3.9), (4.5), and (4.8) not involving w_i are as small as desired, and also so that

$$rcL > d+1. \tag{4.9}$$

Finally, because the interpolation thresholds are centre-independent, the approximations may be selected successively with all w_i so large that

$$\sum_i \frac{1}{w_i}, \qquad \frac{1+\log K}{w_*}$$

are as small as desired. Once all parameters and centres are fixed, H tends to infinity through admissible degrees; then $L_H \to L$, $\delta_H \to 0$, and $(\log M)/H \to 0$.

5 Proof of the conjugate-fibre theorem

Apply Lemma 4.1 at the r embeddings in $\Sigma_\xi(a,b)$ and Lemma 4.2 at the other $d-r$ embeddings. Substitution in (3.10) gives

$$0 \le d(1-\bar\beta_H) + r\max\left\{-cL_H, -\nu\big(A(1-\eta)-\bar\beta_H\big)\right\} + \mathcal{E}_H, \tag{5.1}$$

where, by the choice order above, the nonnegative error $\mathcal{E}_H$ can be made arbitrarily small and then kept fixed while $H\to\infty$.

For the collision alternative,

$$d(1-\bar\beta_H) - rcL_H \le d - rcL_H,$$

which is strictly negative for all sufficiently large H by (4.9). For the large-index alternative, since $r\nu > d$ and $0 \le \bar\beta_H \le \theta$,

$$\begin{aligned} &r\nu\big(A(1-\eta)-\bar\beta_H\big) - d(1-\bar\beta_H) \\ &\quad\ge r\nu\big(A(1-\eta)-\theta\big) - d(1-\theta) = g > 0. \end{aligned}$$

This is exactly the positive gap displayed in (2.4). Thus both entries of the maximum in (5.1), after adding $d(1-\bar\beta_H)$, are bounded above by a fixed negative number. Taking all fixed errors sufficiently small and then H sufficiently large contradicts (5.1).

Therefore (3.1) cannot hold for arbitrarily large denominators when $\nu > 2d/r$. This proves the asserted eventual lower bound and hence $\mu(\xi) \le 2d/r$. As noted in the introduction, ξ is transcendental, and if $r = d$, Dirichlet's theorem supplies the opposite inequality $\mu(\xi) \ge 2$. This completes the proof of Theorem 1.1.

Remark 5.1 (Why the scalar norm threshold is $2d/r$). The constraint $A^2 < \theta < A < 1$ implies

$$1-\theta-2(A-\theta) = 1-2A+\theta > (1-A)^2 > 0.$$

Hence

$$\frac{1-\theta}{A-\theta} > 2.$$

The leading high-index inequality $r\nu(A-\theta) > d(1-\theta)$ therefore forces $\nu > 2d/r$. Conversely, Lemma 2.2 constructs admissible parameters for every strict inequality $\nu > 2d/r$. Thus $2d/r$ is the sharp threshold of this scalar product-formula architecture, although it need not be the true irrationality exponent when $r < d$.

Remark 5.2 (Effectivity). The theorem is qualitative. As in [5], after the centres have been selected the analytic contradiction has an explicit large-degree threshold, but the centre-dependent Serre-vanishing threshold in the interpolation theorem is not made effective.

6 Saturation and optimization

The certificate $e^{a\xi} = b$ is not canonical. Powers, products, and integer combinations may enlarge the proportion of compatible embeddings or reduce the field degree. The following corollaries make that optimization intrinsic.

6.1 Optimization over all certificates

Let

$$\mathcal{C}_{\exp}(\xi) = \{(a,b) \in \overline{\mathbb{Q}}^\times \times \overline{\mathbb{Q}}^\times : e^{a\xi} = b\}.$$

For $(a,b) \in \mathcal{C}_{\exp}(\xi)$, write

$$\rho_\xi(a,b) = \frac{|\Sigma_\xi(a,b)|}{[\mathbb{Q}(a,b):\mathbb{Q}]}.$$

If $\mathcal{C}_{\exp}(\xi) \neq \varnothing$, define the *exponential conjugate-fibre density*

$$\rho_{\exp}(\xi) = \sup_{(a,b)\in\mathcal{C}_{\exp}(\xi)} \rho_\xi(a,b). \tag{6.1}$$

Corollary 6.1 (Optimized conjugate-fibre bound)**.** *If $\mathcal{C}_{\exp}(\xi) \neq \varnothing$, then*

$$\boxed{\mu(\xi) \le \frac{2}{\rho_{\exp}(\xi)}}.$$

In particular, if $\rho_{\exp}(\xi) = 1$, then $\mu(\xi) = 2$.

Proof. Apply Theorem 1.1 to every certificate and take the infimum of the bounds $2/\rho_\xi(a,b)$. □

6.2 Power saturation

Fix one certificate $e^{a\xi} = b$. For $N \ge 1$, put

$$F_N = \mathbb{Q}(a, b^N), \qquad d_N = [F_N : \mathbb{Q}], \tag{6.2}$$

and

$$r_N = \#\left\{\sigma : F_N \hookrightarrow \mathbb{C} : e^{N\sigma(a)\xi} = \sigma(b^N)\right\}. \tag{6.3}$$

Corollary 6.2 (Power saturation)**.** *For every certificate $e^{a\xi} = b$,*

$$\boxed{\mu(\xi) \le 2 \inf_{N\ge1} \frac{d_N}{r_N} = \frac{2}{\sup_{N\ge1}(r_N/d_N)}}.$$

If $r_N = d_N$ for some N, then $\mu(\xi) = 2$.

Proof. For every $N \ge 1$,

$$e^{(Na)\xi} = b^N,$$

and $\mathbb{Q}(Na, b^N) = F_N$. Apply Theorem 1.1 and optimize over N. □

Powering may help in two ways: it can identify conjugates of b that differ only by torsion, and it can replace $\mathbb{Q}(a,b)$ by the smaller field $\mathbb{Q}(a,b^N)$.

Corollary 6.3 (Torsion-defect criterion)**.** *Let L be a normal closure of $\mathbb{Q}(a,b)$ over $\mathbb{Q}$. Suppose that for every embedding $\tau : L \hookrightarrow \mathbb{C}$ the defect*

$$\delta_\tau = \frac{e^{\tau(a)\xi}}{\tau(b)} \tag{6.4}$$

is a root of unity. Then

$$\boxed{\mu(\xi) = 2}.$$

Proof. Choose N divisible by the orders of the finitely many roots of unity δ_τ. Then

$$e^{N\tau(a)\xi} = \tau(b)^N = \tau(b^N)$$

for every τ. Every embedding of $F_N = \mathbb{Q}(a, b^N)$ extends to an embedding of L, so $r_N = d_N$. Apply Corollary 6.2. □

6.3 The relation lattice

Suppose that

$$e^{a_j\xi} = b_j, \qquad a_j, b_j \in \overline{\mathbb{Q}}^\times, \qquad 1 \le j \le s.$$

For $n = (n_1, \dots, n_s) \in \mathbb{Z}^s$, set

$$A_n = \sum_{j=1}^{s} n_j a_j, \qquad B_n = \prod_{j=1}^{s} b_j^{n_j}. \tag{6.5}$$

Whenever $A_n \neq 0$, we have $e^{A_n\xi} = B_n$. Put

$$F_n = \mathbb{Q}(A_n, B_n), \qquad d_n = [F_n : \mathbb{Q}],$$

and

$$r_n = \#\left\{\sigma : F_n \hookrightarrow \mathbb{C} : e^{\sigma(A_n)\xi} = \sigma(B_n)\right\}.$$

Corollary 6.4 (Relation-lattice saturation)**.** *With the notation above,*

$$\boxed{\mu(\xi) \le 2 \inf_{\substack{n \in \mathbb{Z}^s \\ A_n \neq 0}} \frac{d_n}{r_n}.}$$

In particular, an integer combination whose conjugate fibres are all compatible proves $\mu(\xi) = 2$, even when none of the original certificates has full compatibility.

Proof. Apply Theorem 1.1 to the certificate $e^{A_n\xi} = B_n$ and optimize over n. □

In a common normal field, define the individual defects

$$\delta_{\tau,j} = \frac{e^{\tau(a_j)\xi}}{\tau(b_j)}.$$

The combined certificate is compatible at τ precisely when

$$\prod_{j=1}^{s} \delta_{\tau,j}^{n_j} = 1.$$

When the defects are roots of unity, the search for a fully compatible combination is therefore a finite system of linear congruences.

6.4 Positive algebraic logarithms

For $\alpha \in \overline{\mathbb{Q}}_{>0} \setminus \{1\}$, define its *power degree* by

$$d_{\mathrm{pow}}(\alpha) = \min_{N \geq 1} [\mathbb{Q}(\alpha^N) : \mathbb{Q}]. \tag{6.6}$$

Corollary 6.5 (Algebraic logarithms)**.** *For every positive algebraic $\alpha \neq 1$,*

$$\boxed{\mu(\log \alpha) \leq 2 d_{\mathrm{pow}}(\alpha).}$$

If $\alpha^N \in \mathbb{Q}$ for some $N \geq 1$, then

$$\boxed{\mu(\log \alpha) = 2}.$$

Proof. Use the powered certificate

$$e^{N \log \alpha} = \alpha^N.$$

In the field $\mathbb{Q}(\alpha^N)$, exactly one embedding fixes the generator α^N, so $r_N = 1$. Thus Corollary 6.2 gives

$$\mu(\log \alpha) \leq 2[\mathbb{Q}(\alpha^N) : \mathbb{Q}]$$

for every N. Minimize over N. If one power is rational, the upper bound is 2, and Dirichlet gives equality. The rational-power case is also obtained directly in [5, Corollary 7]. □

7 Examples

Corollary 7.1. *For every positive rational D,*

$$\mu(\pi\sqrt{D}) = \mu\left(\frac{\pi}{\sqrt{D}}\right) = 2.$$

Proof. For $\xi = \pi\sqrt{D}$, use $a = i/\sqrt{D}$ and $b = -1$; for $\xi = \pi/\sqrt{D}$, use $a = i\sqrt{D}$ and $b = -1$. Every embedding sends a to a or $-a$, and $e^{\pm i\pi} = -1$, so all fibres are compatible. □

Taking $D = 1$ recovers $\mu(\pi) = 2$, the theorem of [7].

Corollary 7.2. *For every nonzero rational number t,*

$$\boxed{\mu(\arctan t) = 2}.$$

Proof. Let $\xi = \arctan t$. Then

$$e^{2i\xi} = \frac{1 + it}{1 - it} \in \mathbb{Q}(i).$$

Complex conjugation sends $2i$ to $-2i$ and the right-hand side to its inverse, so both embeddings of $\mathbb{Q}(i)$ are compatible. □

Corollary 7.3 (Quadratic units)**.** *Let $K = \mathbb{Q}(\sqrt{D})$ be a real quadratic field, and let $\varepsilon > 0$ be a unit. If $\mathrm{N}_{K/\mathbb{Q}}(\varepsilon) = 1$, then*

$$\mu\left(\frac{\log \varepsilon}{\sqrt{D}}\right) = 2.$$

The same conclusion holds for norm -1 after replacing the certificate by its square.

Proof. Put $\xi = \log \varepsilon/\sqrt{D}$, $a = \sqrt{D}$, and $b = \varepsilon$. For norm 1, the nontrivial automorphism sends $a \mapsto -a$ and $b \mapsto b^{-1}$, so both fibres are compatible. For norm -1, square the relation and apply Corollary 6.2. □

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The geometric interpolation theorem and the analytic row-translation and collision estimates used here are those of Liu–Jiang–Zhang [5], building on the interpolation-determinantmethod of Laurent [3, 4] and the recent proof for π [7]. The new step in this note is the simultaneous number-field denominator clearing and product-formula comparison across compatible and incompatible conjugate fibres.

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